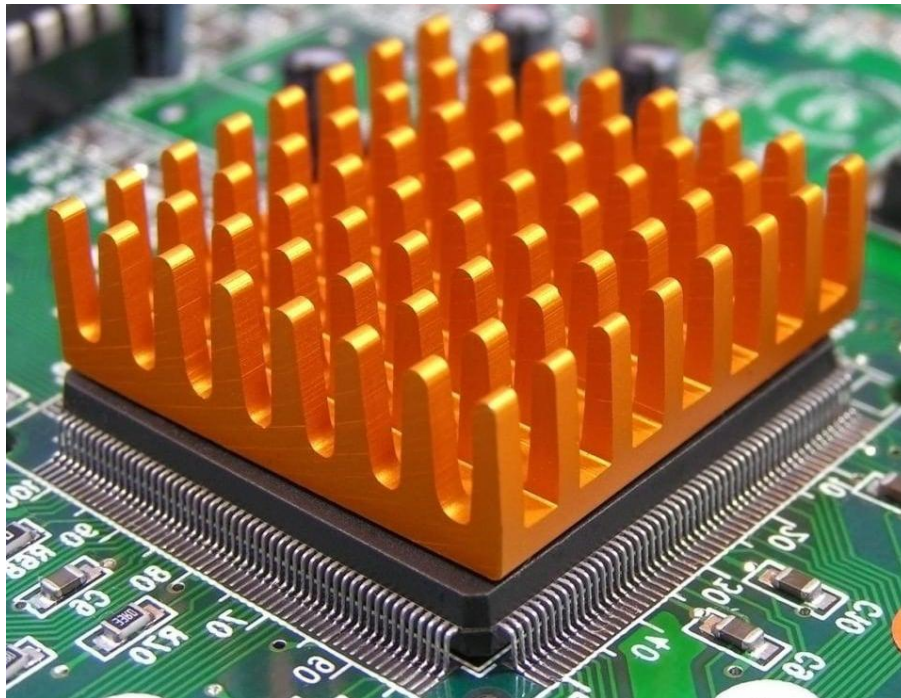


ME4429 - Thermofluid Application and Design
Design & Optimization Through Thermal Analysis of Heat Sinks
Team 6 - Samuel Kelly and Giovanni Crema
02/07/2025



Author Introduction

The authors of this paper are junior mechanical engineering major students pursuing their bachelor's and master's degrees together in 4 years.

Samuel Kelly is a born and raised Canadian who moved to America for school. He chose to be a mechanical engineer because of his deep passion for cars, innovation and CAD. Sam's goal is to one day become an entrepreneur of an engineering consulting firm that sells innovative technologies and software to EPC companies.

Giovanni Crema was born in Italy and moved to the United States when he was younger. He chose to become a mechanical engineer after playing with legos, learning math, and building his own speaker systems when he was younger. Gio is interested in working in the medical product industry one day and owning his own restaurant. This summer he will be taking on an internship at Philips Medisize and working as a Manufacturing Engineer.

1.0 - Problem Statement and Project Scope

As new and innovative computer hardware is being developed there is an extremely high demand for effective thermal management to accommodate these systems. If heat production from hardware is not contained it can cause a significant loss of performance and lifespan (Bergman et al., 2017). Our focus today is on computer chips and dissipating their heat production using heat sinks. Heat sinks utilize the physics of conduction and convection, by increasing the surface area and heat transfer laws heat sinks can keep computer chips running at an optimal temperature (Incropera et al., 2011). Some key applications of heat sinks are in computer hardware like CPUs

and GPUs. Also spanning further in the automotive industry with use cases on Electric Vehicles to the battery and other electronic systems from overheating.

This project focuses on utilizing SolidWorks to design and optimize a heat sink for computer chips under various conditions. With SolidWorks, each design will be simulated to observe the performance of the heat sink. Focusing specifically on how it performs under different convection conditions, then determining if the design is optimal and if not, adjusting parameters like fin spacing height and thickness. A successful result is a heat sink that maintains the chips within their optimal temperature with a small range in temperature variation.

RESULTS

2.0 - Part 1 – Simulation of chips without the heat sink

Objective

The purpose of this initial simulation is to obtain a baseline of the thermal behaviour of the chips without a heat sink. With this information, we can compare the results of the integration of the heat sink and discover the overall impact.

2.1 Model

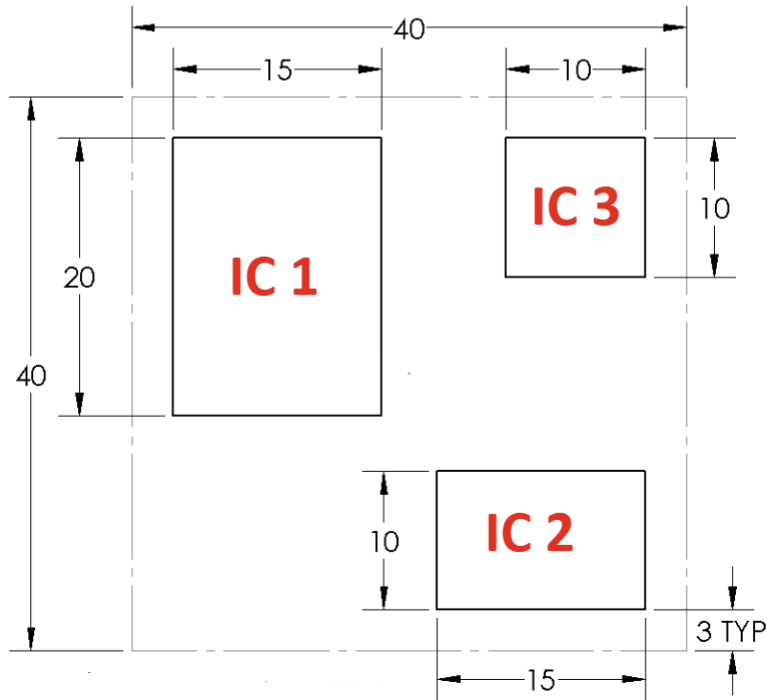
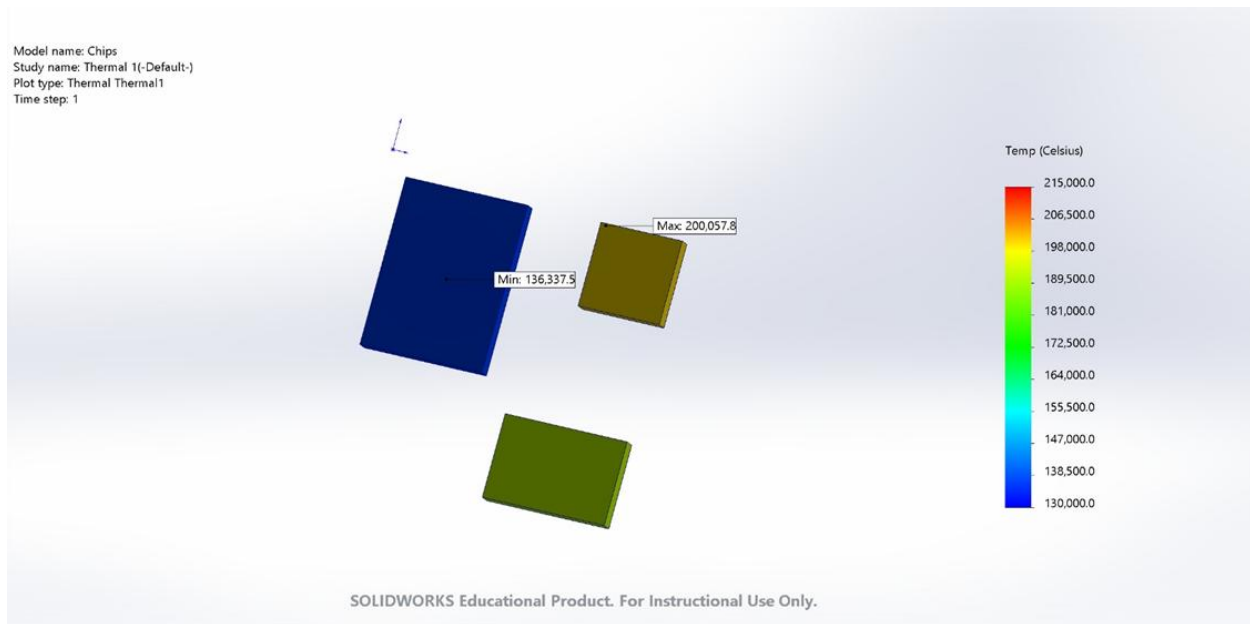


Figure 1: Model of 3 IC chips

Utilizing SolidWorks a simulation was done using steady-state thermal analysis, with units of Kelvin and N/m^2 . The model was comprised of 3 chips with heat generation values of 100W, 80W and 60W. The model had boundary conditions with a natural convection applied with a convection coefficient of $10 \text{ W}/((\text{m}^2)(\text{K}))$ and an ambient temperature of 298 K. The bottom surface was insulated to prevent heat loss through conduction ($q = 0$). The material chosen for the chips was silicon with a thermal conductivity of $0.2 \text{ W}/((\text{m})(\text{k}))$. The simulation generated 14,512 nodes and 8,549 elements, which gives confidence that the mesh is valid.

2.2 Simulation Temperatures



***Figure 2:** The figure above is an image of the three chips at 100 W, 80 W and 60 W. On the right is a temperature gradient that is being displayed by the colour of the chips.*

The simulation provided the following temperatures for the chips:

Chip 1 (100 W): 200,057.8 °C

Chip 2 (80 W): 169,800.4 °C

Chip 3 (60 W): 136,337.5 °C

Conclusion:

It is very easy to see that these temperatures are extremely outside of the hardware's optimal temperature and would cause instant component failure. This proves the necessity of heatsinks for the hardware to properly dissipate the heat generated.

3.0 - Part 2 – Simulation of Chips with the Heat Sink

Objective

The purpose of this simulation is to investigate the performance of a heat sink's thermal management of the three chips. Further, this simulation will be run at three different convection coefficients to determine the minimum coefficient to maintain the temperatures at optimal operational levels.

3.1 Model

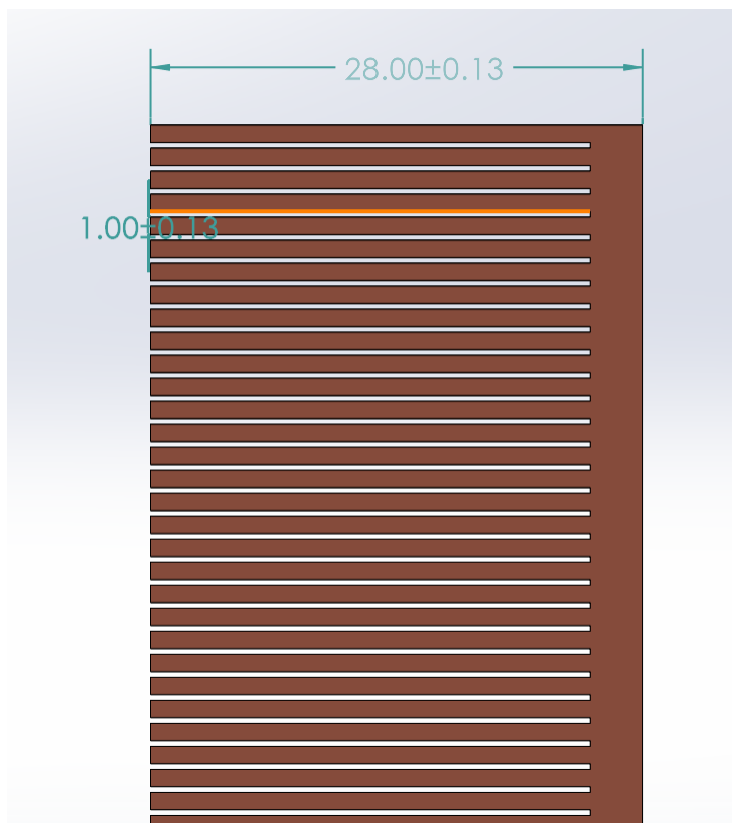


Figure 3: *Image of Heat Sink Design*

Utilizing SolidWorks a simulation was done using steady-state thermal analysis, with SI units of Kelvin and N/m^2 . The model was comprised of 3 chips with heat generation values of 100W, 80W and 60W. The model had boundary conditions with an ambient temperature of 298 K. The

bottom surface was insulated to prevent heat loss through conduction ($q = 0$). The material chosen for the chips was silicon with a thermal conductivity of $0.2 \text{ W}/((\text{m})(\text{k}))$ for the chips.

Further, a heat sink was introduced to dissipate heat and allow the three chips to operate at peak performance while maintaining an acceptable temperature. The heat sink is made of pure copper which has a thermal conductivity of $390 \text{ W}/((\text{m})(\text{k}))$. Copper's value is great for this application and will allow for a more simplistic and minimal fin design.

We are considering three different convection cases both natural and forced:

Case 1: (Low Convection - Natural Convection): $h = 10 \text{ W}/((\text{m}^2)(\text{K}))$

Case 2: (Medium Convection - Moderate Airflow): $h = 50 \text{ W}/((\text{m}^2)(\text{K}))$

Case 3: (High Convection - Strong Forced Cooling): $h = 100 \text{ W}/((\text{m}^2)(\text{K}))$

3.2 Simulation Temperatures

The simulation provided the following temperatures for the chips:

Case 1: (Low Convection - Natural Convection): $h = 10 \text{ W}/((\text{m}^2)(\text{K}))$

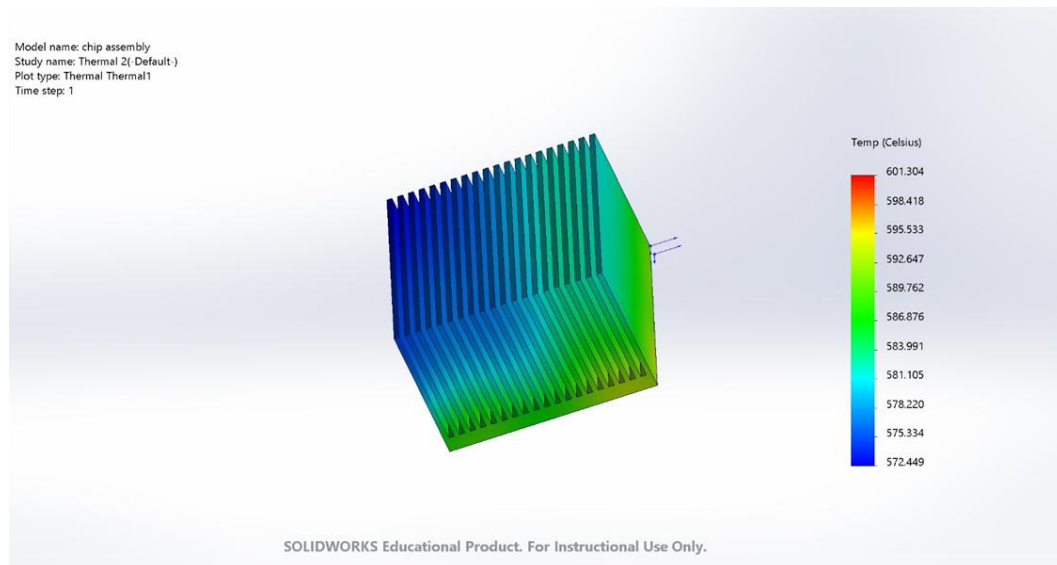


Figure 4: This image shows the temperature distribution of the heat sink at low convection (Front View).

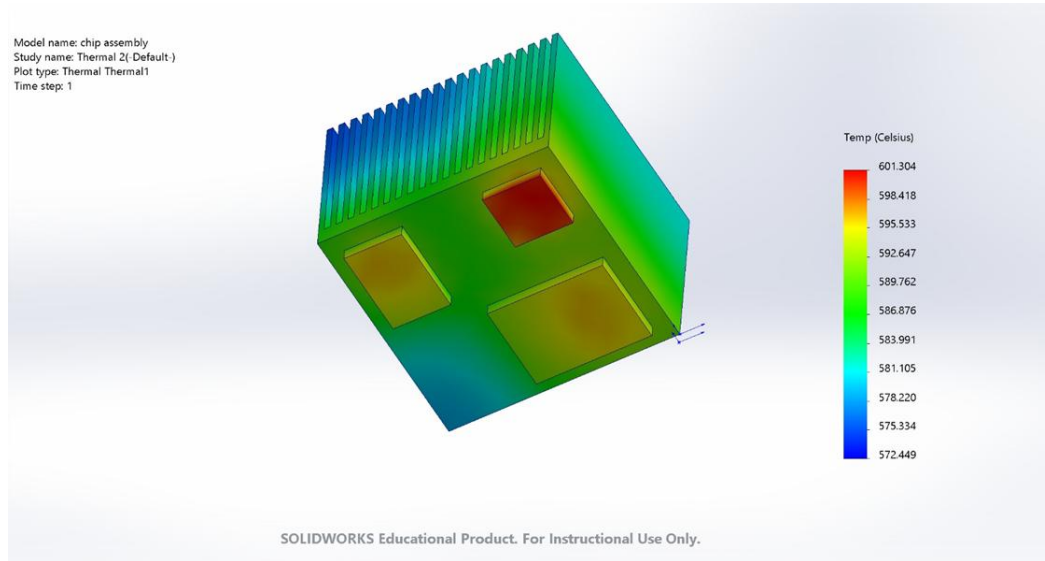


Figure 5: Results from Case 1, Bottom View

The results from the first case are visible in the above figures. The maximum temperature is shown on the heat chips, which is expected as they are the heat source. The maximum temperature on chip 3 (60 W) is 601.3°C, far beyond the acceptable operational temperature.

Case 2: (Medium Convection - Moderate Airflow): $h = 50 \text{ W}/((\text{m}^2)(\text{K}))$

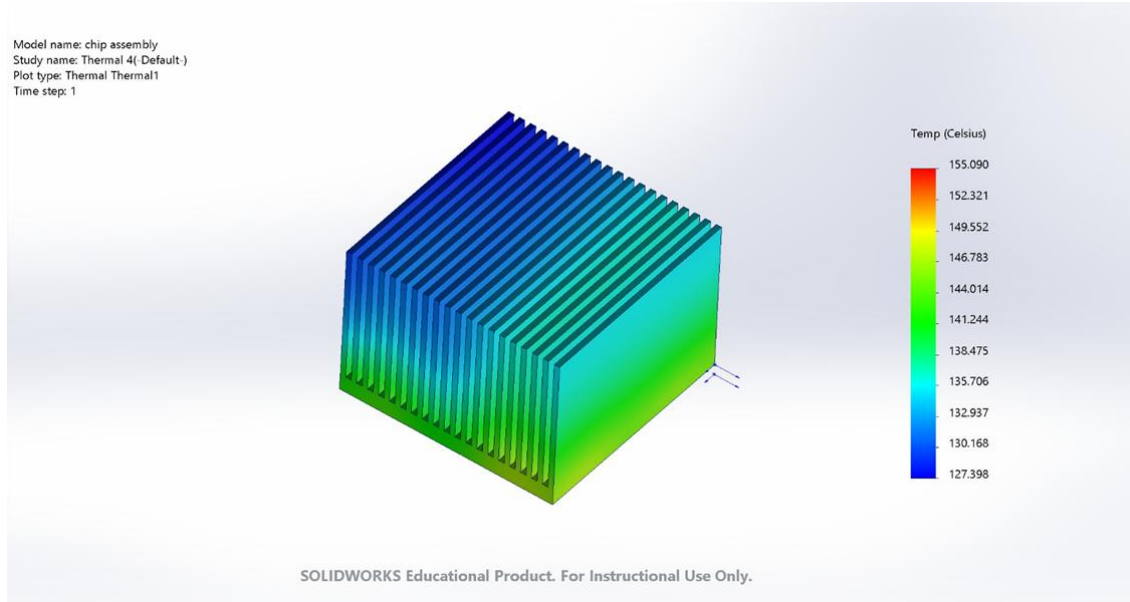


Figure 6: *Front view of temperature distribution of the heat sink at medium convection.*

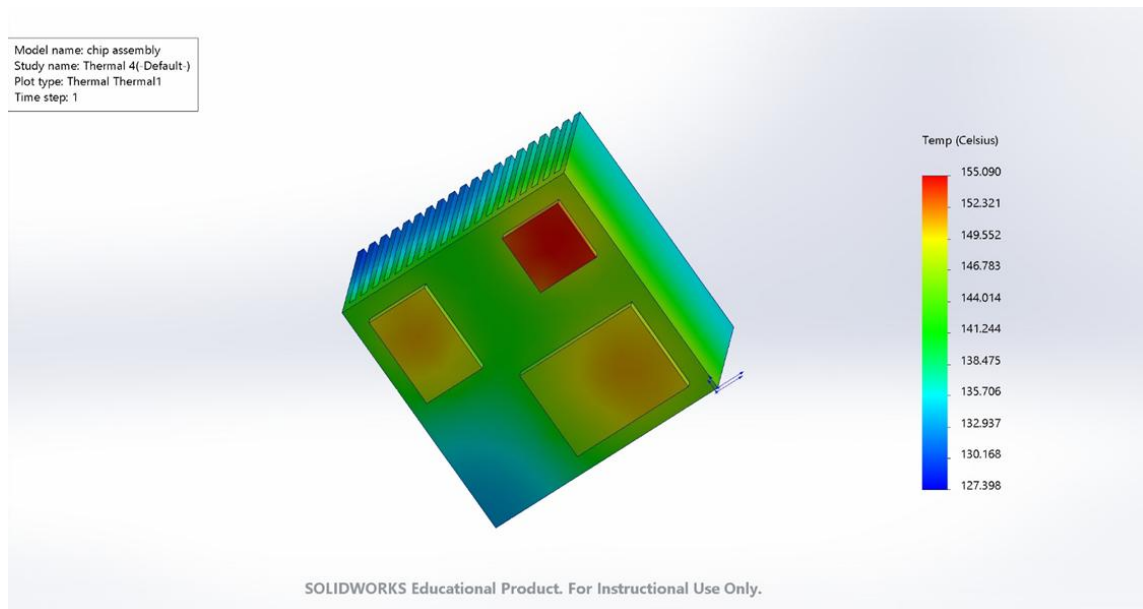


Figure 7: *Case 2, Bottom View*

The second case's results are shown above and like the first case the maximum temperature is on-chip 3 (60 W). The highest temperature is significantly lower than the first case at 155.1°C however this is still beyond the operational temperature.

Case 3: (High Convection - Strong Forced Cooling): $h = 100 \text{ W}/((\text{m}^2)(\text{K}))$

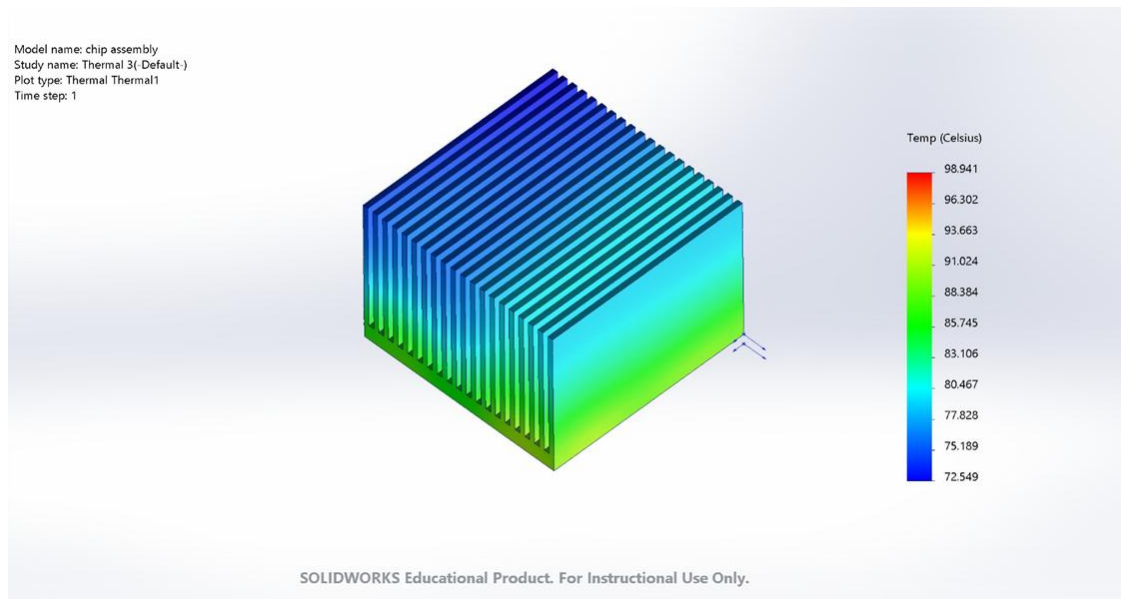


Figure 8: *Front view of Case 3 shows the temperature distribution of the heat sink at high forced convection.*

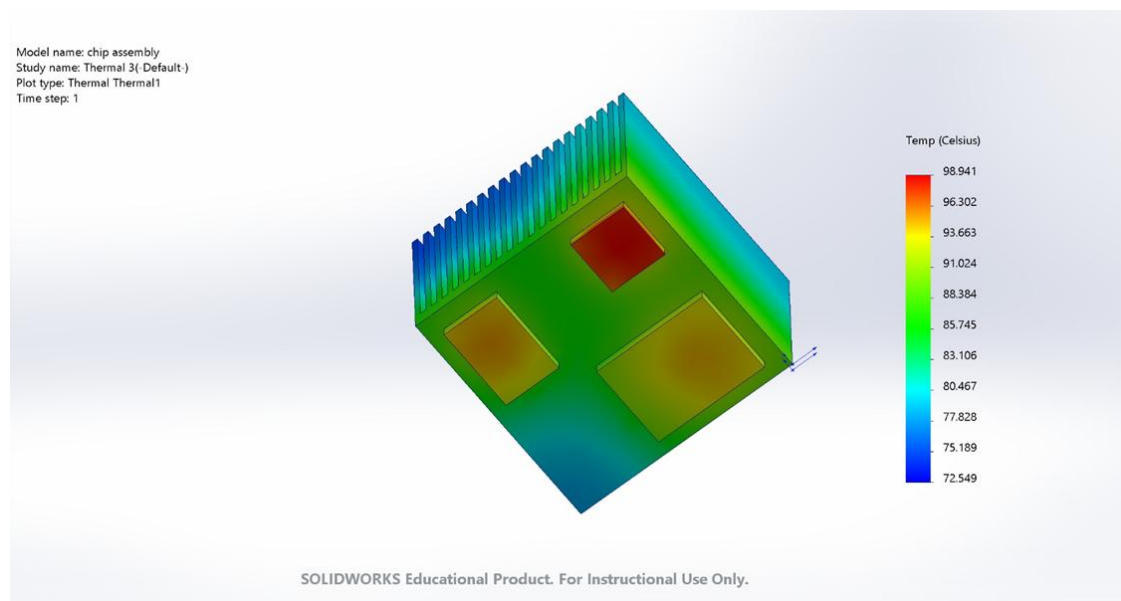


Figure 9: *Case 3, Bottom view*

The third case comes very close to operational temperature with the maximum temperature on chip 3 at 98.9°C.

The introduction of a heat sink to the chips had a significant impact on thermal management decreasing the temperature by hundreds of thousands of degrees. In part 1 chip 3 experienced a temperature of 136,337.5 °C this is how extensive the chips are without anything to assist in heat dissipation. With the integration of the heat sink chip 3 was significantly the highest in all connection simulations peaking with an h value of 10 (W)/((m²)(K)) at 127.4°C with $h = 50$ (W)/((m²)(K)), and finally 72.5°C with $h = 100$ (W)/((m²)(K)). The forced convection had a great impact on the temperature however chip three remained higher than the rest indicating that there is still a challenge with it for the future.

3.3 Top of Fins Temperature Distribution

In this subsection, different portions of the heat sink's temperature distribution are observed, shown in the two figures below. Figure 10 shows the position of the selected nodes and Figure 11 displays a plot of the temperatures over time. The graph confirms the hypothesis of higher temperatures nearest to the chips and back plate.

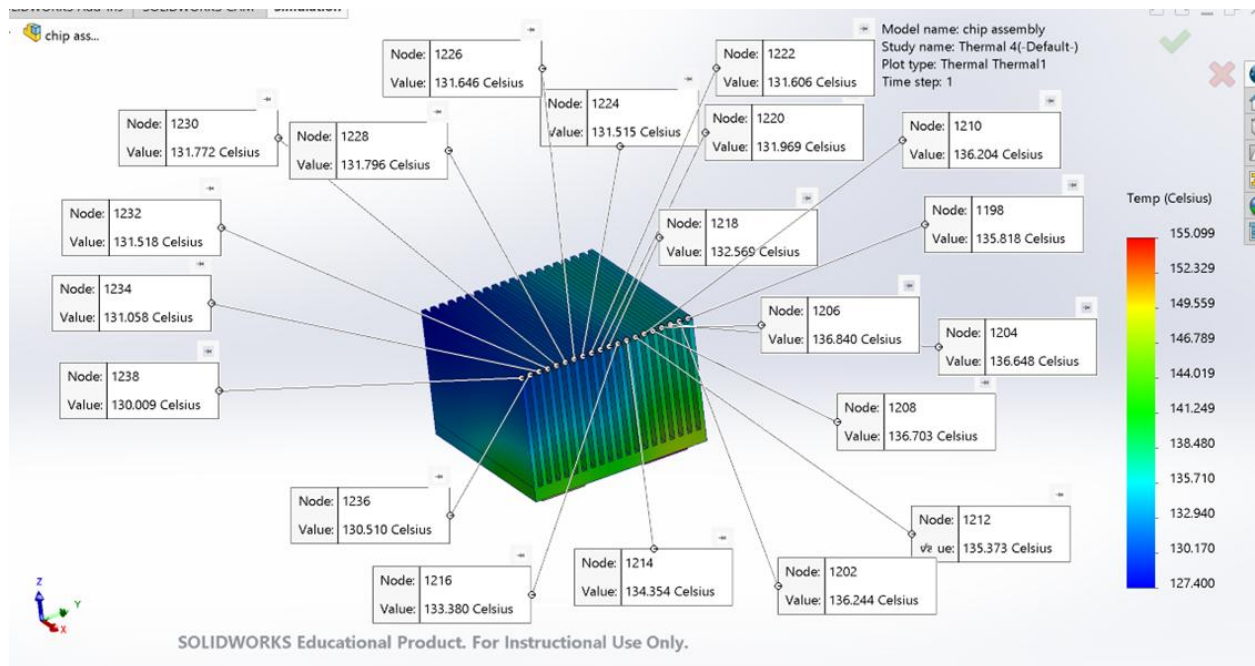


Figure 10: Positions of nodes across the top of the heat sink and their corresponding temperatures.

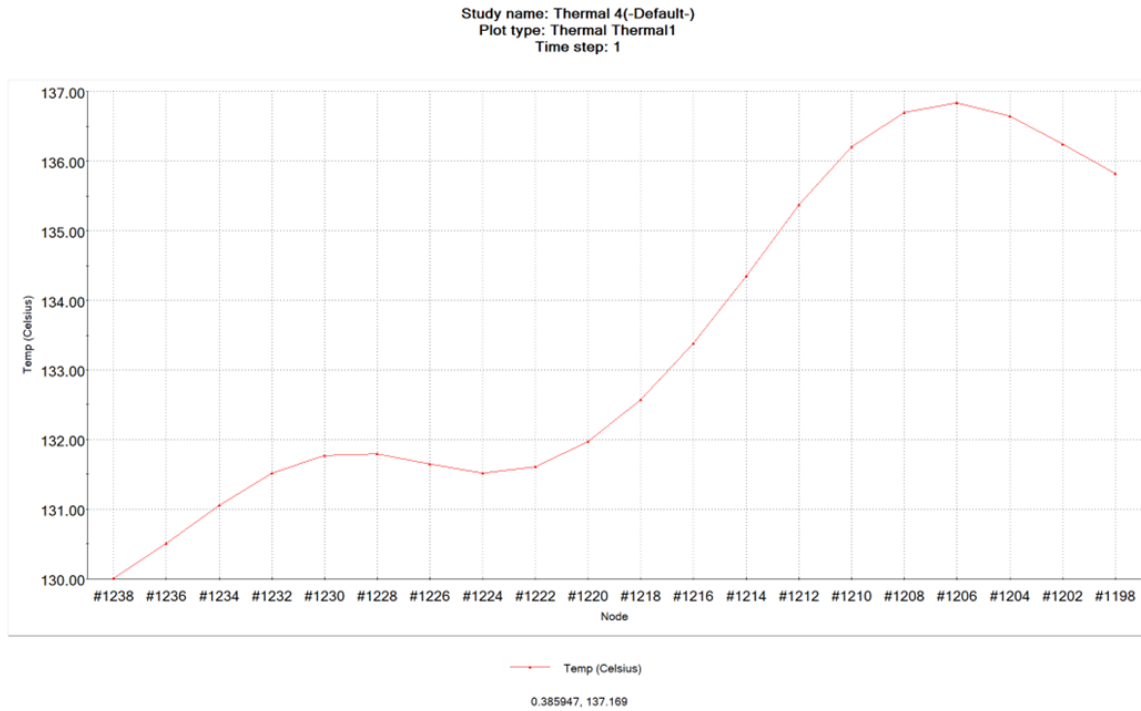


Figure 11: *Plot of temperatures at each node across the top of the heat sink.*

3.4 Temperature Distribution along Vertical edge of fin

In this subsection, we identified the temperature distributions at different nodes along one of the edges of the vertical fins on the heatsink. The temperature distribution is as we predicted, the farther the node is from the source of heat, the lower the temperature is.

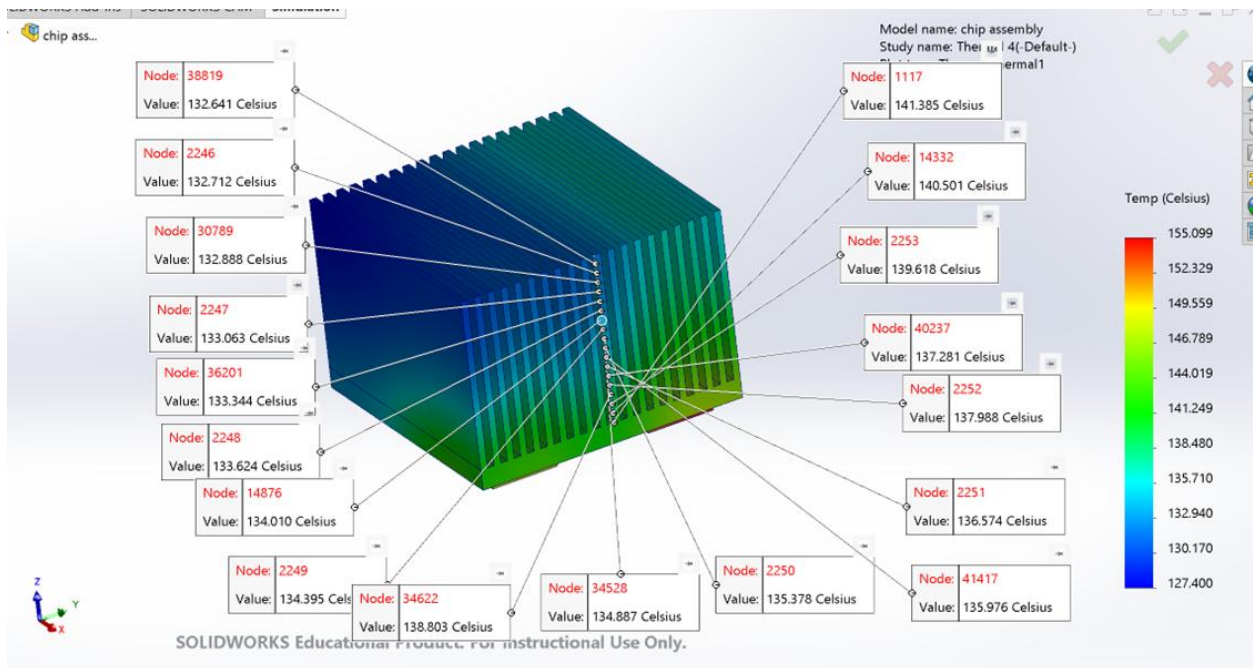


Figure 12: Positions of nodes along one vertical fin of the heat sink and their corresponding temperatures.

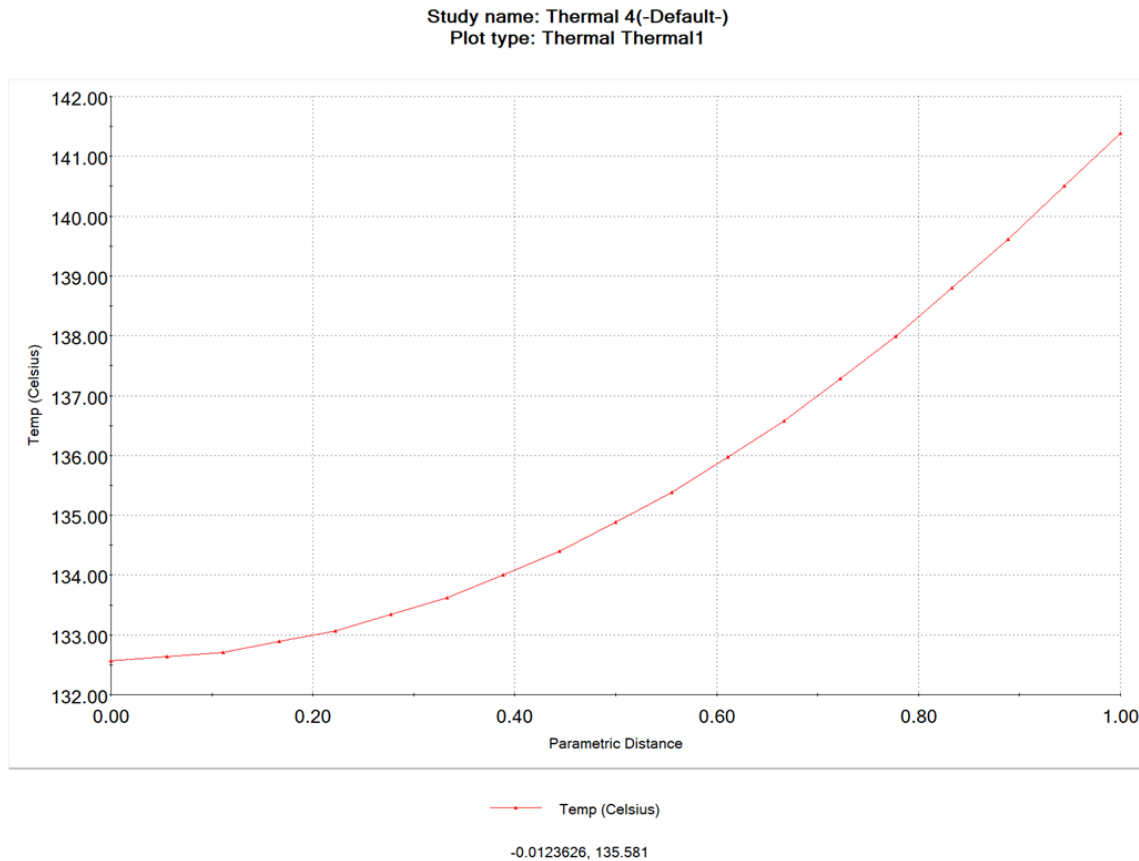


Figure 13: *Plot of temperatures at each node along one vertical fin of the heat sink*

Conclusion:

The temperature results are as expected with the temperature and convection coefficient being inversely proportional proving the necessary innovation of cooling systems in computers today. The natural convection less to extreme overheating of the chips with all-surpassing 570°C . The moderate case made substantial improvements from the natural case however, still fell short of what is acceptable. The forced cooling with an h value of $100(\text{W}/((\text{m}^2)(\text{K})))$ had great thermal management maintaining all chips under 100°C . This result proves the need for cooling devices like fans or forced airflow.

4.0 - Part 3 – Simulation of the Heat Sink in a Transient Condition

Objective

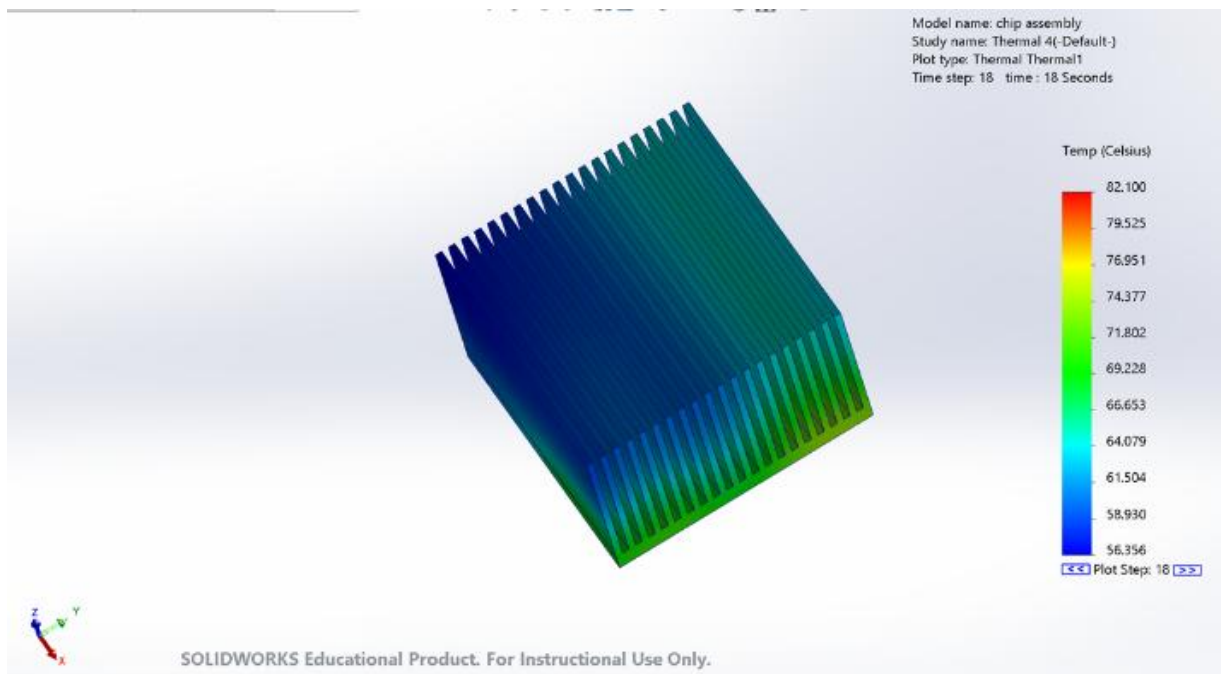
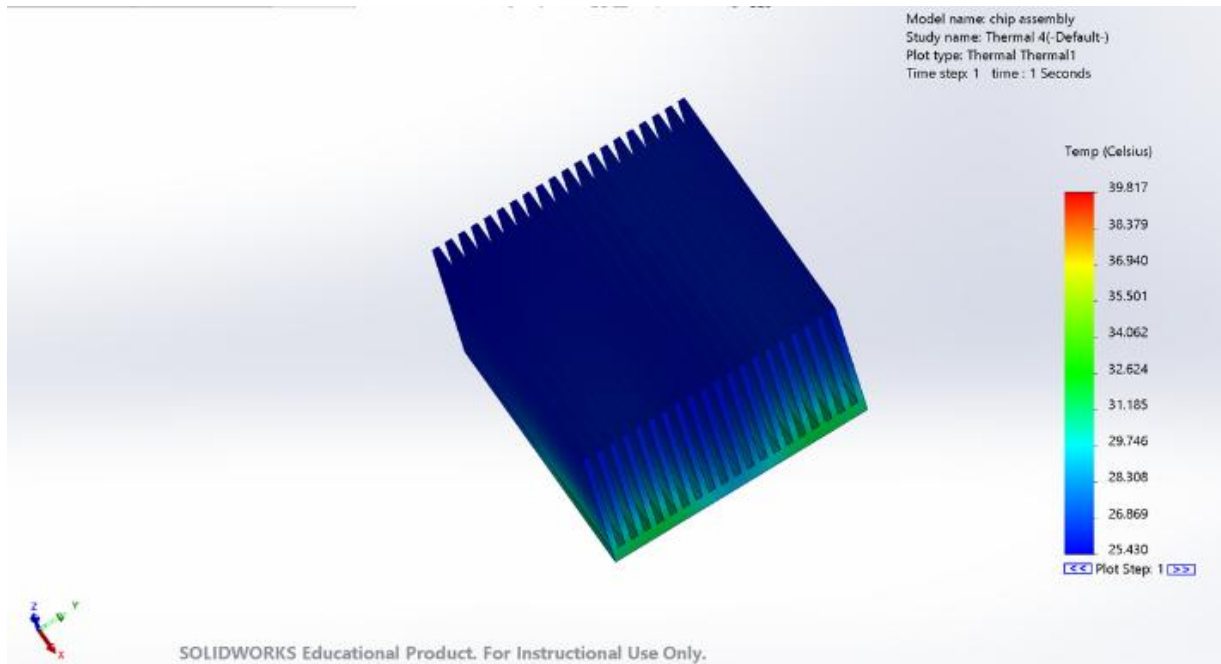
The purpose of this initial simulation is to obtain a baseline of the thermal behaviour of the chips without a heat sink. With this information, we can compare the results of the integration of the heat sink and discover the overall impact.

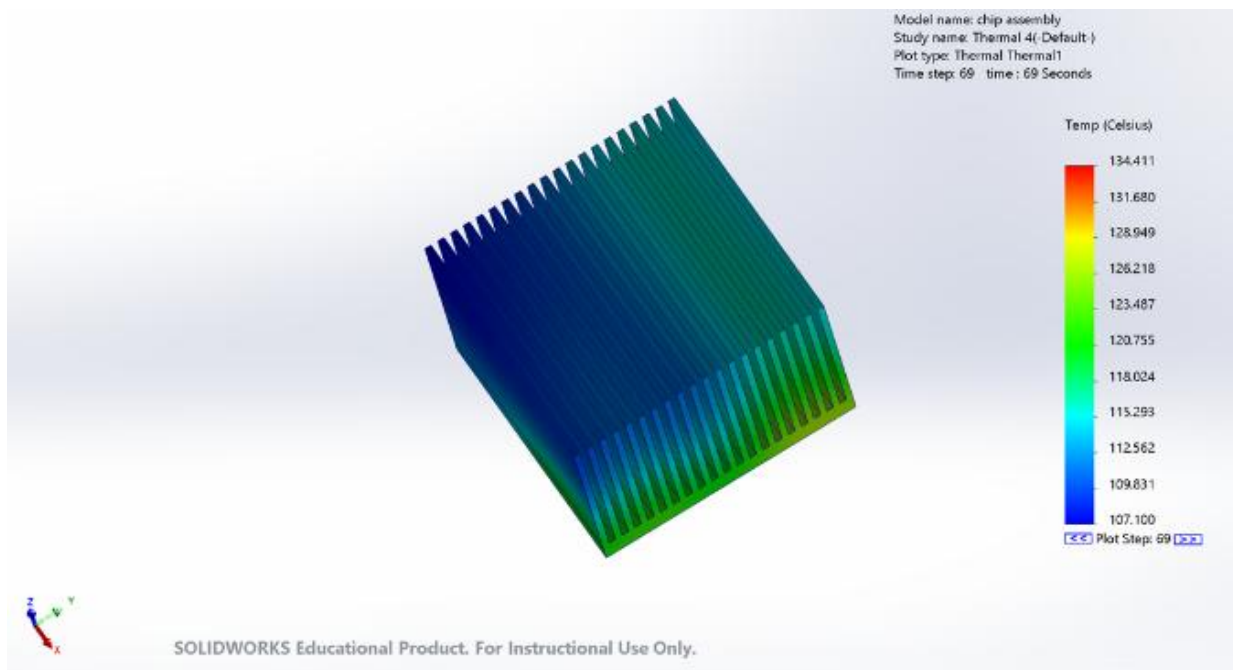
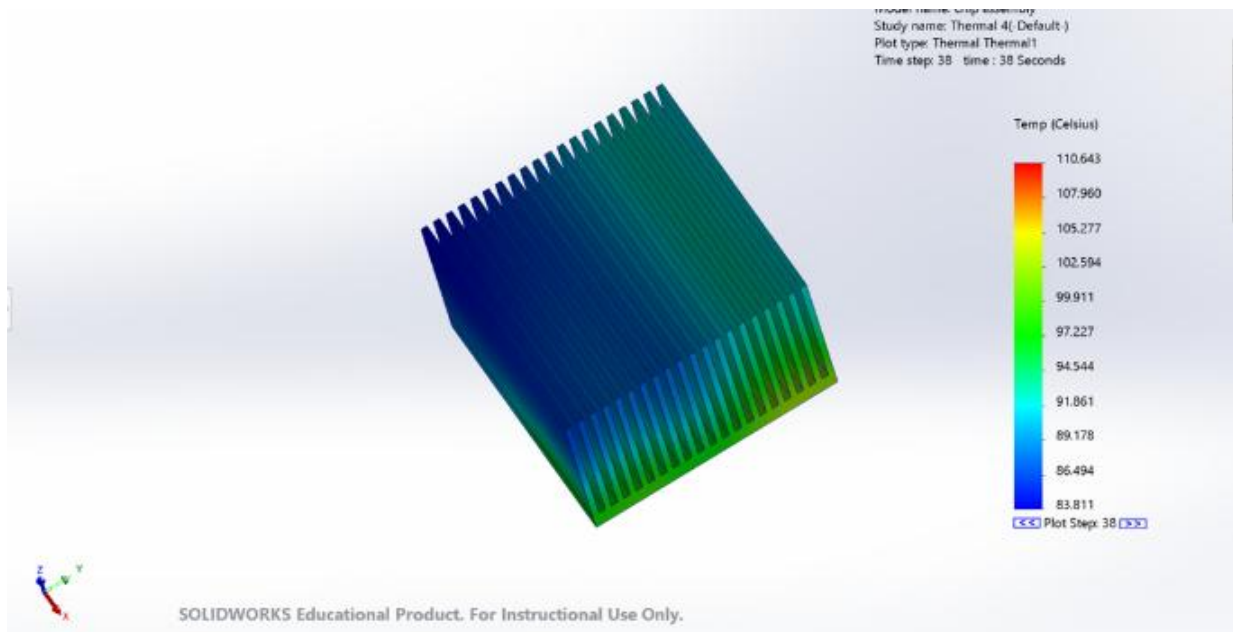
4.1 Model

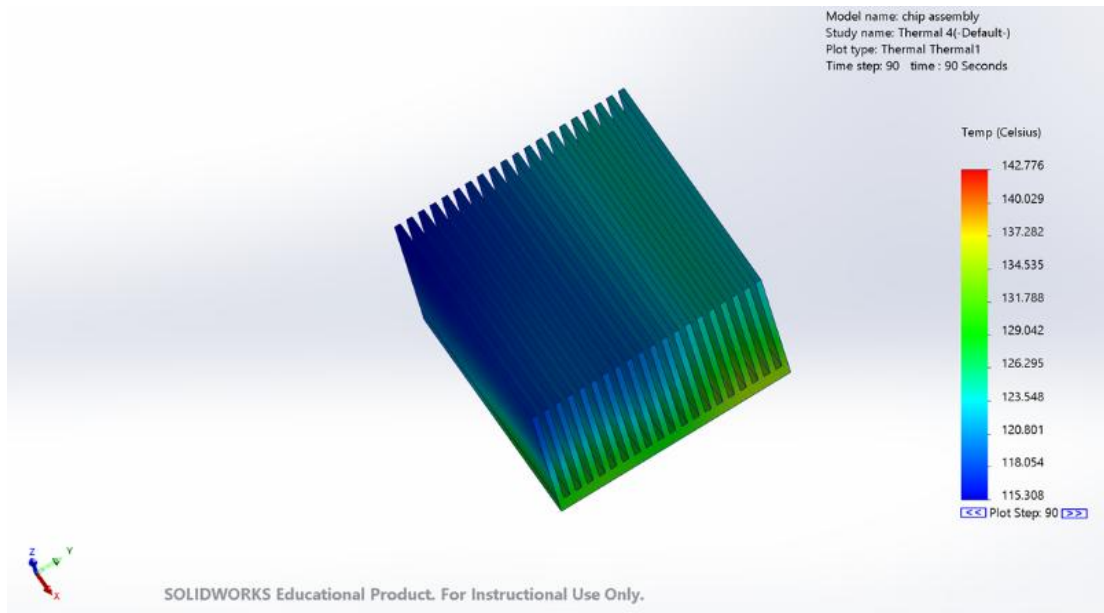
Utilizing SolidWorks a simulation was done using steady-state thermal analysis, with units of Kelvin and N/m^2 . The model was comprised of 3 chips with heat generation values of 100W, 80W and 60W. The model had boundary conditions with an ambient temperature of 298 K. The bottom surface was insulated to prevent heat loss through conduction ($q = 0$). The material chosen for the chips was silicon with a thermal conductivity of 0.2 W/((m)(K)) for the chips.

Further, a heat sink was introduced to dissipate heat and allow the three chips to operate at peak performance while maintaining an acceptable temperature. The heat sink is made of pure copper which has a thermal conductivity of 390 W/((m)(K)) . Copper's value is great for this application and will allow for a more simplistic and minimal fin design. A convection coefficient of $h = 50 \text{ (W/((m}^2\text{)(K))}$ simulating moderate convection and cooling was used.

4.2 Simulation of Temperatures on Heat Sink in a transient condition







Figures 14-18: *A view of the distribution of the transient thermal simulation of the heat sink at the various time steps $t=1s$ to $t=90s$.*

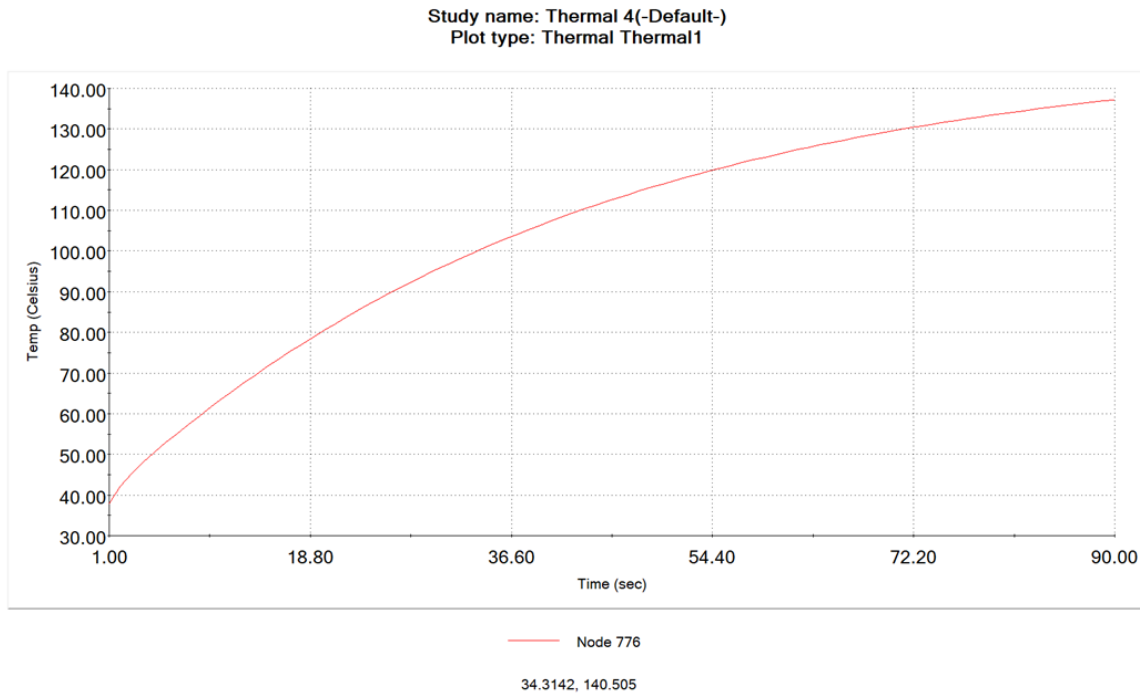


Figure 19: *Plot of temperatures on a point of IC 2, with each time step along a period of 90s.*

t = 1s: At the start of the transient simulation, the temperature was at approximately 313 K and the temperature gradients were noticeable and the chips were quickly rising in temperature, although the heat sink was slowing the increase by conducting the heat away.

t = 19s: The heat sink was finally distributing heat evenly among the fins, and the temperature began to plateau just above 353 K.

t = 38 seconds: This is where the system reached a temperature of approximately 110°C, this was representative of an effective heat sink however not under optimal temperatures.

t = 69 seconds: Temperature at 134 °C.

t = 90 seconds: Temperature at 143 degrees °C continuing to rise and will plateau eventually and reach a steady state.

Conclusion:

It is very easy to see that these temperatures are extremely outside of the hardware's optimal temperature and would cause instant component failure. This proves the necessity of heatsinks for the hardware to properly dissipate the heat generated.

5.0 - Part 4 – Finding an acceptable fin configuration

In this section, the objective is to optimize the heat sink design by adjusting the fin configuration and overall design to get the temperature of all three chips below 85°C. Similar boundary conditions from part two are being used for this simulation. Three designs are below, an under design that does not meet the acceptable temperature, an overdesign that works better than needed and a design that reaches the perfect acceptable temperature. by iterating through these designs the most efficient configuration can be determined for cooling while minimizing size.

5.1 Case 1: Under design

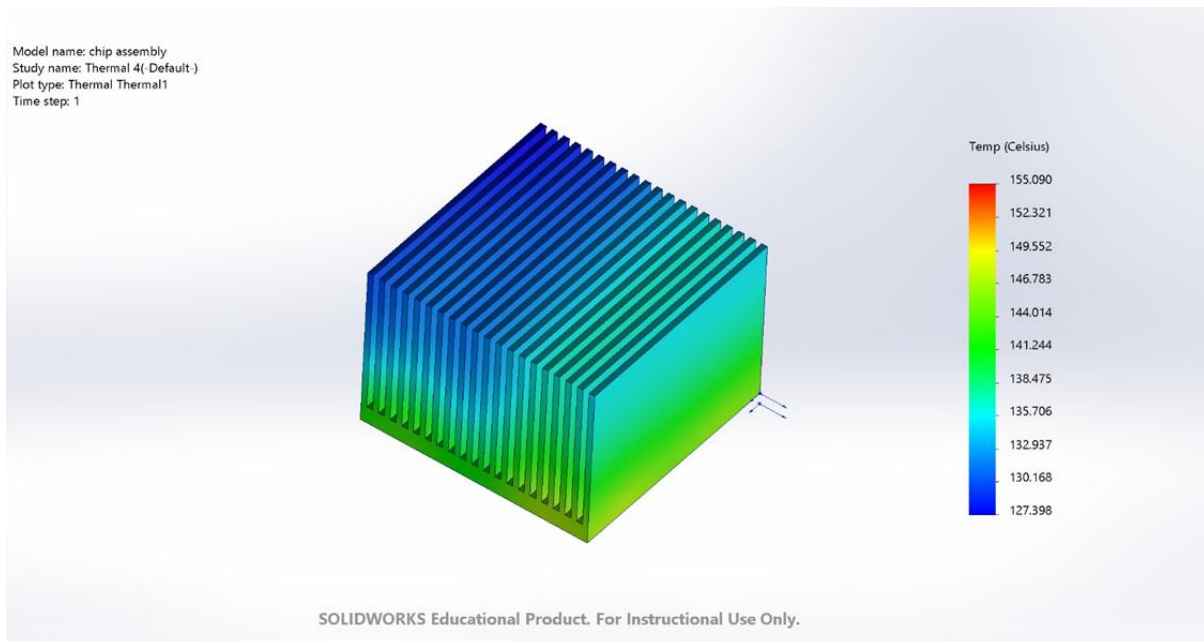


Figure 20: *Temperature distribution of the under designed heat sink, front view.*

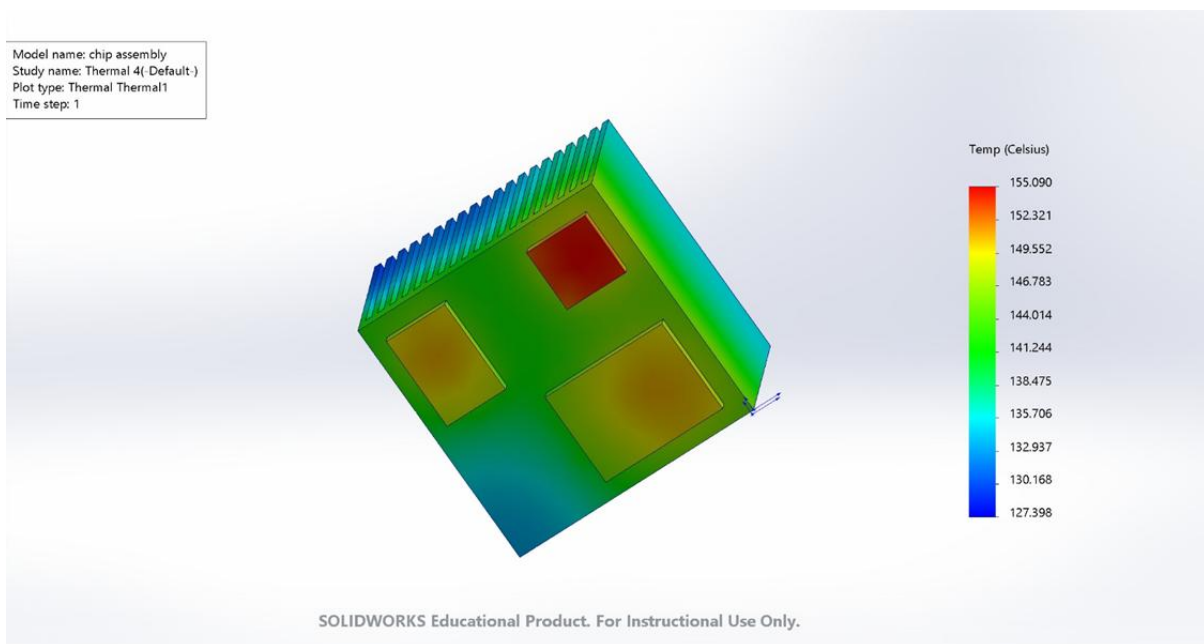


Figure 21: *Temperature distribution of the under designed heat sink, bottom view.*

5.2 Case 2: Acceptable Design

To create an acceptable design for the heatsink to maintain a temperature at 85 degrees Celsius, we decided to expand the surface area of the bottom of the heat sink. Having a greater surface area, allows the chips to dissipate heat along a larger space through their conductive properties.

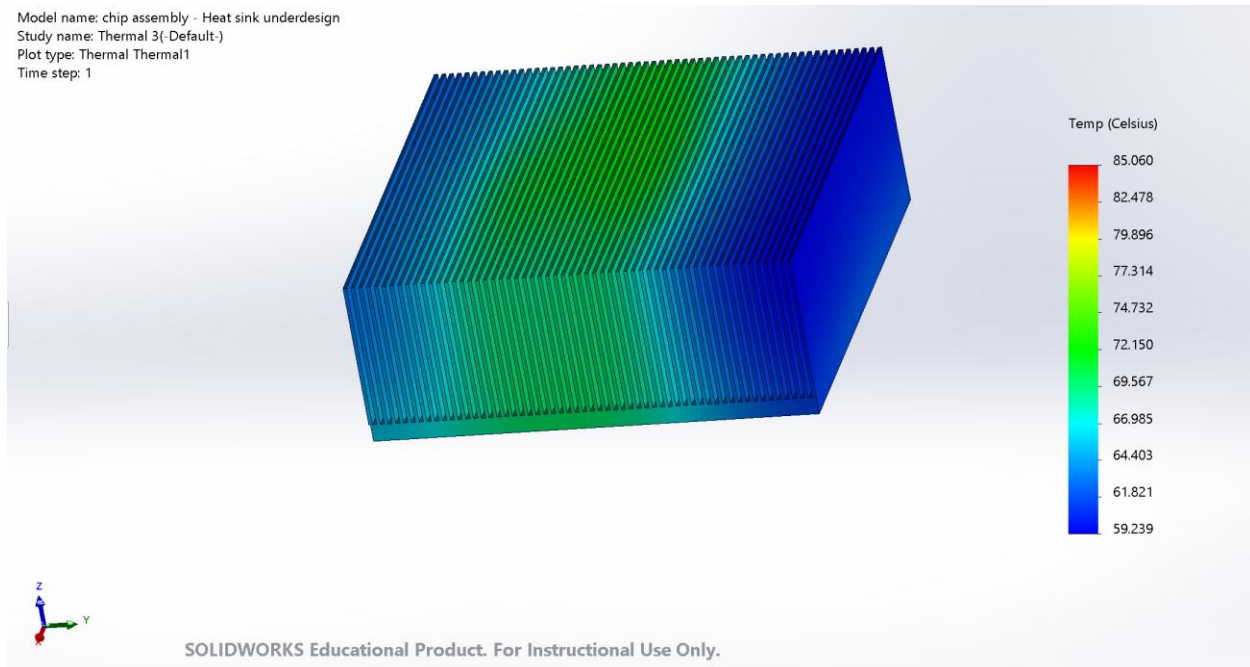


Figure 22: *Temperature distribution of an acceptable design, front view*

5.3 Case 3: Over Designed

To create an over designed heatsink to have a temperature at below 85 degrees Celsius, we decided to expand the surface area of the bottom of the heat sink, increase the fin length and decrease fin width which maximizes the surface area to volume ratio and therefore allows the chips to dissipate heat along a larger space through their conductive properties.

Model name: chip assembly - Heat sink underdesign
Study name: Thermal 3(-Default-)
Plot type: Thermal Thermal1
Time step: 1

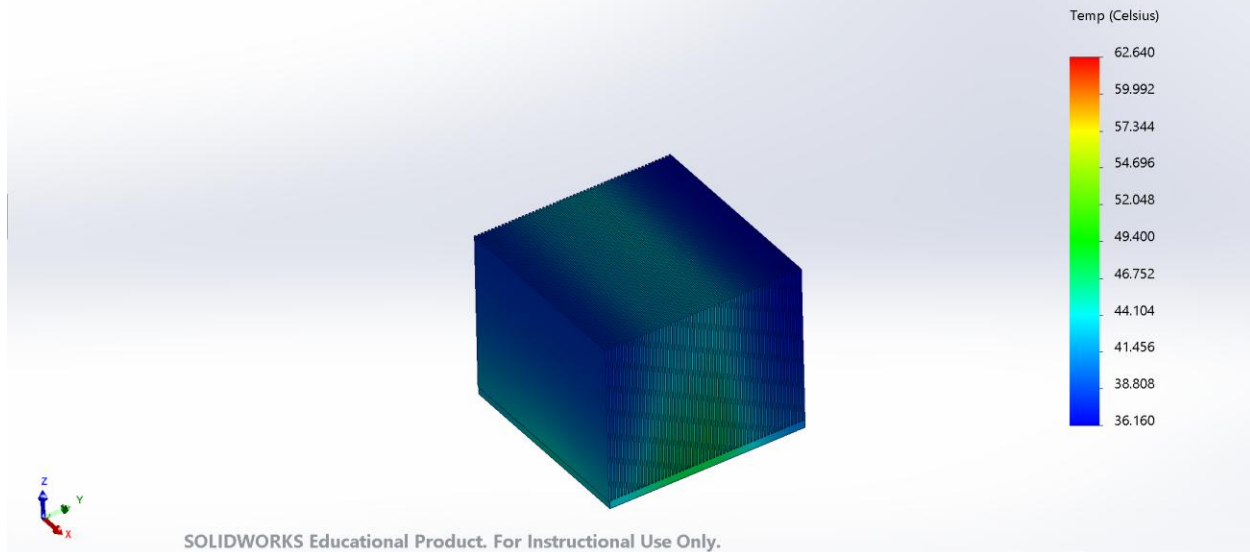


Figure 23: *Temperature distribution of an over designed heat sink, front view*

Model name: chip assembly - Heat sink underdesign
Study name: Thermal 3(-Default-)
Plot type: Thermal Thermal1
Time step: 1

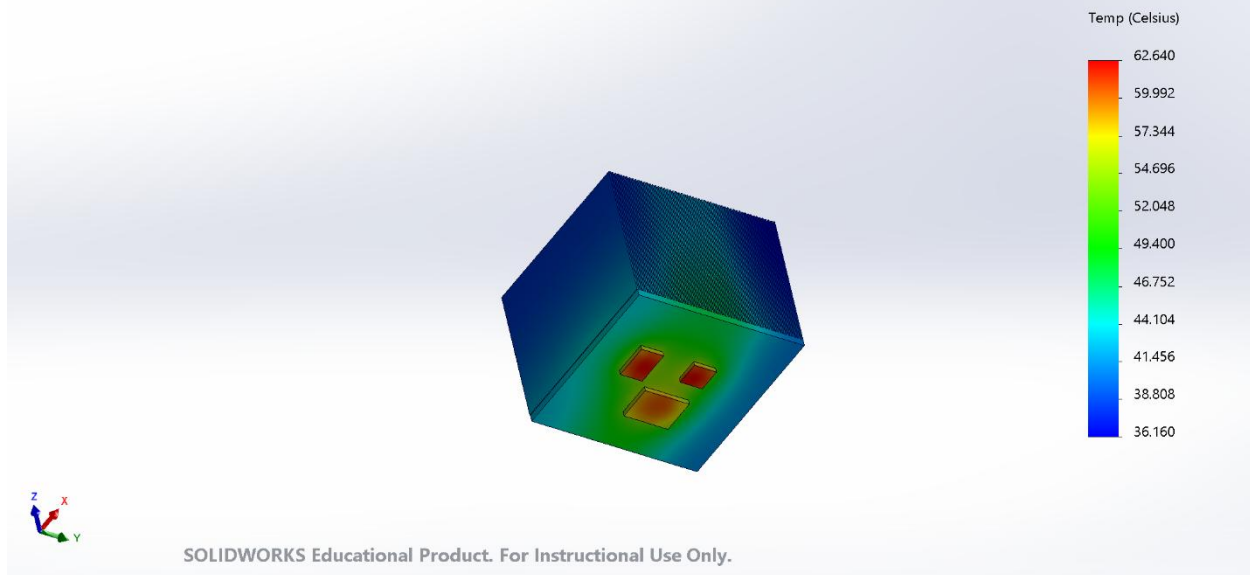


Figure 24: *Temperature distribution of an over designed heat sink, bottom view*

6.1 Conclusion

This project aided in furthering the understanding of the principles of heat transfer and thermal management, building upon the concepts learned in ES3001. In ES3001 the focus was on the theory of conduction, convection and radiation. This project applied the learnings of ES3001 by integrating real-world applications through the use of SolidWorks. SolidWorks allowed for visualization of the numerical analysis of temperature distributions of the chips and heat sink system. A significant takeaway from this study is that thermal management is detrimental to the success of systems that generate heat. Simply allowing natural convection to run its course does not do enough to contain the chip's temperature at an operational level. It can be noted that high forced convection however does have a large impact. The transient analysis enabled an analysis of the systems performance over time, which showed that a heat sink is not an instant fix and rather something that needs time to work. It operates by dissipating heat gradually through the principles and physics learned in ES3001.

References:

1. Incropera, F. P., DeWitt, D. P., Bergman, T. L., & Lavine, A. S. (2011). *Fundamentals of Heat and Mass Transfer*. Wiley.
2. [Bergman, T. L., Lavine, A. S., Incropera, F. P., & DeWitt, D. P. \(2017\). *Introduction to Heat Transfer*. Wiley.](#)

